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(71)Name of Applicant :

1)RAMCO INSTITUTE OF TECHNOLOGY

Address of Applicant :KRISHNAPURAM PANCHAYAT NORTH VENGANALLUR VILLAGE RAJAPALAYAM-626117 -----

Name of Applicant : NA

Address of Applicant : NA

(72)Name of Inventor :

1)K.Jeyageetha

Address of Applicant :Department of CSE, North Venganallur Village, Rajapalayam, Tamil Nadu India, 626117 -----

2)G.Sakthi Priya

Address of Applicant :Department of CSE, North Venganallur Village, Rajapalayam, Tamil Nadu India, 626117 -----

3)P.Devisri

Address of Applicant :Department of CSE, North Venganallur Village, Rajapalayam, Tamil Nadu India, 626117 -----

4)S.Manjula

Address of Applicant :Department of CSE, North Venganallur Village, Rajapalayam, Tamil Nadu India, 626117 -----

5)Dr.K.Vijayalakshmi

Address of Applicant :Department of CSE, North Venganallur Village, Rajapalayam, Tamil Nadu India, 626117 -----

(57) Abstract :

ABSTRACT In poultry fanning, the control of gas emissions from litter-primarily ammonia (NH₃) and hydrogen sulfide (H₂S)--is critical for safeguarding environmental quality and ensuring the health of poultry and workers. This invention introduces an integrated automated system that incorporates gas sensors, a microcontroller unit (NodeMCU ESP8266), ventilation fans, a conveyor belt, an alarm system, and a pesticide light to manage gas emissions in real time. Upon detecting gas concentrations above safety thresholds, the system autonomously triggers alarms [6], activates the conveyor belt for litter removal [4], and engages fans to ventilate the area [3]. A UV-based pesticide light also supports hygiene and pest control. The system supports remote monitoring and control via the Blynk IoT platform [5], allowing farm-specific customization of threshold settings and mitigation strategies. The proposed solution demonstrates a cost-effective, scalable, and sustainable approach to emission management in poultry environments [1][2], addressing key limitations in existing fragmented or manual systems [3].

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